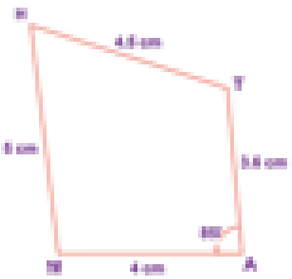
5.10.3 Construct a quadrilateral when its 4 sides and one angle are given

Example 5.23

Construct a quadrilateral *MATH* with *MA*=4 *cm*, *AT*= 3.6 *cm*, *TH* = 4.5 *cm*, *MH*= 5 *cm* and $\angle A = 85°$. Also find its area.

Solution:

Given:
MA=4 *cm*, *AT*= 3.6 *cm*,
TH = 4.5 *cm*, *MH*= 5 *cm* and $\angle A = 85°$



Rough diagram

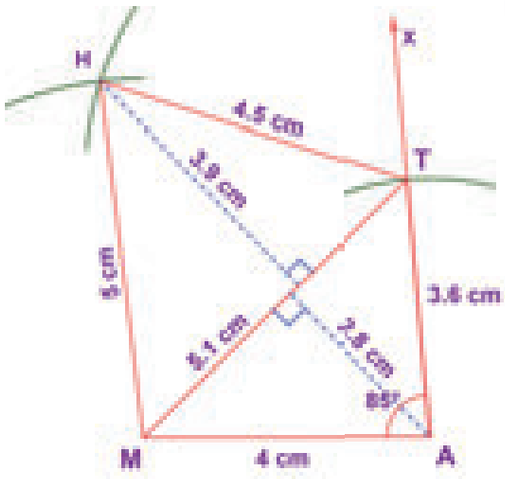


Fig. 5.42

Steps:

1. Draw a line segment *MA* = 4 *cm*.
2. Make $\angle A = 85°$.
3. With *A* as centre, draw an arc of radius 3.6 *cm*. Let it cut the ray *AX* at *T*.
4. With *M* and *T* as centres, draw arcs of radii 5 *cm* and 4.5 *cm* respectively and let them cut at *H*.
5. Join *MH* and *TH*.
6. *MATH* is the required quadrilateral.

Calculation of Area:

Area of the quadrilateral *MATH* $= \frac{1}{2} \times d \times (h_1 + h_2)$ *sq. units*

$= \frac{1}{2} \times 5.1 \times (3.9 + 2.8)$

$= 2.55 \times 6.7 = 17.09\ cm^2$

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